

Team Marula — Impact Statement

Our project, “Optimising RAG Document Retrieval for Agronomic Advice,” is an agricultural information retrieval system designed to help farmers and agricultural extension workers find relevant agronomic information more efficiently. Farmers often need reliable information about crops, pests, diseases, and farming practices, but relevant knowledge can be difficult to locate among large amounts of agricultural information. Our system addresses this problem by retrieving and ranking the most relevant agricultural documents for a given farming-related query.

The project could positively impact smallholder farmers, agricultural extension workers, and other people who depend on timely agricultural information. By helping users locate relevant documents more quickly, the system could reduce the time and effort required to search for useful knowledge. This is important because better access to relevant information can support more informed farming decisions and potentially contribute to improved productivity, sustainable agricultural practices, and community food security. Organisations involved in agricultural extension and farmer support could also benefit from more efficient access to existing knowledge resources.

However, the system also presents risks. The most significant risk is that retrieved information may be incomplete, outdated, poorly suited to a particular location, crop, or farming condition, or ranked highly despite not being the most appropriate source. If users treat retrieved information as universally applicable, this could contribute to poor farming decisions. Other risks include regional and cultural underrepresentation in the documents, unequal representation of crops and farming systems, and potential limitations caused by language or digital access.

To manage these risks, our team will prioritise reliable and diverse agricultural sources, retain source and licensing information, and make the origin of retrieved information traceable. We will evaluate retrieval performance using relevance judgements and monitor gaps in the representation of crops, countries, and farming contexts. The system should be treated as a support tool rather than a replacement for agricultural experts or extension officers. User and stakeholder feedback can also help identify missing information and improve the system over time.

Our project promotes fairness, transparency, inclusivity, autonomy, and community well-being by aiming to make existing agricultural knowledge easier to discover while recognising that farmers have different needs and circumstances. At the same time, our impact assessment has limitations. Retrieval performance does not necessarily mean that the information is correct or that it will improve real-world farming outcomes. Our current dataset may not fully represent all farmers, regions, languages, or local agricultural practices. Therefore, further testing with diverse users and agricultural stakeholders would be necessary before making broader claims about the project's impact.